ConvoShield

Redefining Women's Digital Safety with Cognitive AI

GDG CODE4HER HACKATHON 2026 SUBMISSION • TRACK: CYBERSECURITY

01 The Digital Crisis

In the age of hyper-connectivity, women are increasingly targeted by predatory psychological tactics including **Love Bombing, Sextortion, and Financial Coercion**. Traditional security tools fail because they scan for *malicious links*, not *malicious behavior*.

02 How ConvoShield Protects

Real-Time Watchdog

Continuous WhatsApp Web monitoring via our specialized Chrome Extension that detects and alerts on suspicious chat spikes.

Vision Analytics

Proprietary OCR system (Tesseract.js) allowing users to analyze screenshots directly, bypassing manual data entry.

Cognitive Engine

Deep-learning analysis of conversation tropes using NVIDIA's nim for low-latency, high-accuracy threat detection.

Emergency Loop

Immediate linkage to National Cyber Cell 1930 and gender-specific support networks integrated into the analysis UI.

03 Technical Architecture

FRONTEND
React + Vite

→

MIDDLEWARE
Chrome Ext API

→

AI INFERENCE
NVIDIA NIM

→

EXPORT
Safety PDF

Total Privacy: No chat logs are permanently stored on servers. Data is processed through an ephemeral secured pipeline.

04 Targeted Impact

We designed ConvoShield to cater to specific high-risk personas identified in cybercrime reports:

The Young Professional

Targeted via LinkedIn or dating apps for professional/romantic scams. ConvoShield alerts her to "over-familiarity" red flags.

Elderly Women

Frequently targeted by "relative in distress" or parcel scams via WhatsApp calls. ConvoShield analyzes the urgency patterns.

05 Security Library & Encyclopedia

Empowerment through Awareness

ConvoShield doesn't just scan; it teaches. Our integrated **Security Library** provides a real-time feed of trending Indian scams (Digital Arrest, AI Voice Clones) ensuring users are one step ahead of predators.

06 Scalability & Future Roadmap

- ✔ **Multilingual Support:** Expanding analysis to Hindi, Marathi, and Tamil to protect non-English speaking demographics.
- ✔ **Voice Analysis:** Integrating audio-to-text to scan suspicious voice notes for emotional manipulation.
- ✔ **Community Reporting:** An anonymous portal to report new scam tropes to improve our global detection model.
- ✔ **Corporate Integration:** Deploying ConvoShield into HR platforms to prevent workplace digital harassment.

Judge's Summary

"ConvoShield AI is a comprehensive safety ecosystem built with a human-centric approach. By combining real-time extension monitoring, advanced OCR, and the NVIDIA NIM inferencing engine, we provide a complete protective layer for women navigating the digital world."